Eight Big Questions About AI

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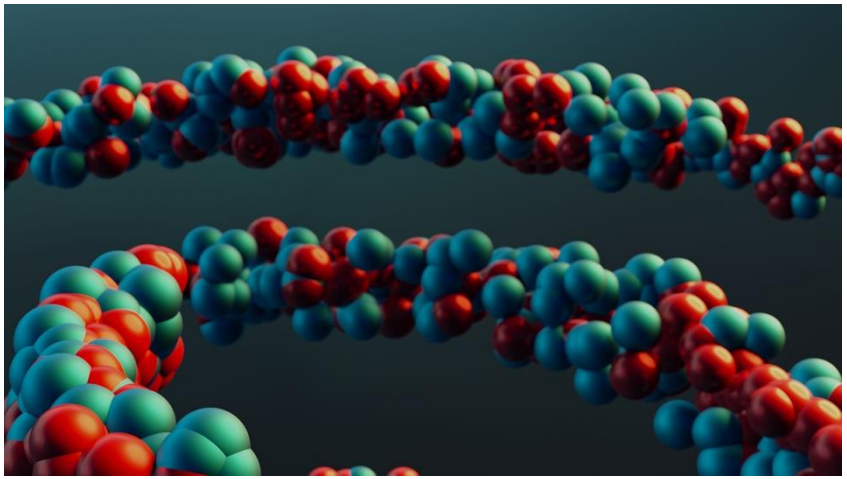
Based on discussions by Terence Tao

- And my own research
- Webinar — November 6, 2026
- Mark Kerzner

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1. Models

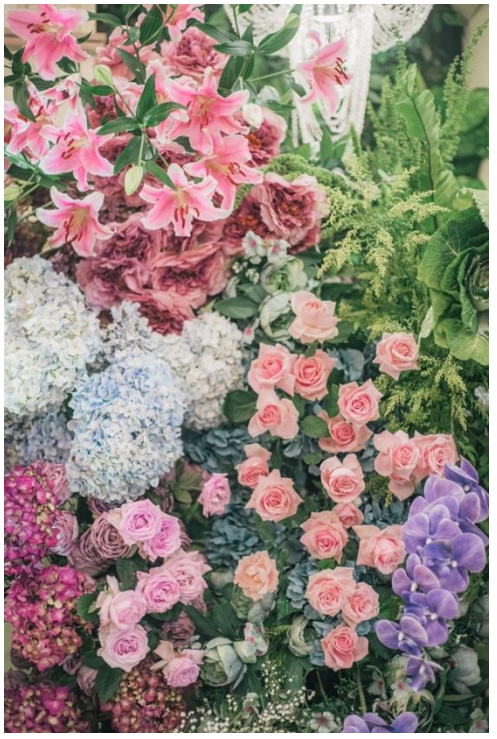
- Models in physics describe the world
  - Outside its assumptions, a useful model becomes misleading
  - But physics knows where its model stops
- Finance did not — the crisis of 2008
  - For hedging, different model used assumptions that were true in crisis



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2. AI is also a model

- We designed the training — we do not understand **what** the models learned
  - "Roses are red, and violets are..."
  - Models sound intelligent, but do they really **understand** ?



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3. Why don’t we know where AI's limits lie?

- Tests reveal particular successes and failures — not the boundaries of understanding
- Does a failed test show a real limit, or only this model's?
- We need much more mathematical work to map it



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4. Is AI good for mathematics?

- Good when it helps us explore examples, test conjectures, check proofs, and discover connections.
- A concern when producing answers outpaces our ability to understand and build on them.
- Chess comparison: machines are stronger, and people still play.
  - But chess has fixed rules and a winner; mathematics has neither.



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5. Why may AI be bad for mathematics?

- The proof is the fruit — new concepts, methods and connections are the tree
- Picking the fruit does not tend the tree
- So who is tending it? - **us**
- But leaving the problem unsolved grows nothing either



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6. In what way is building AI like raising a child?

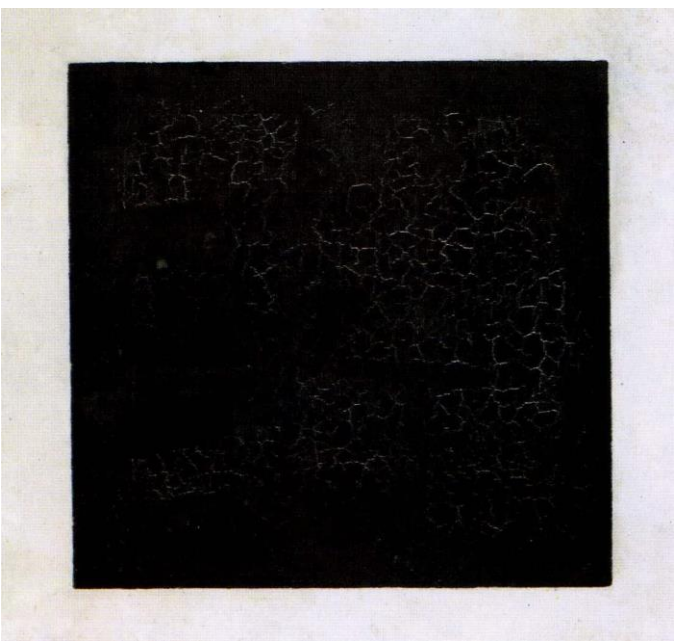
- We teach by examples, feedback and boundaries
- But we cannot specify every situation in advance
- You do not know how he or she will turn out



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7. Why is AI dangerous?

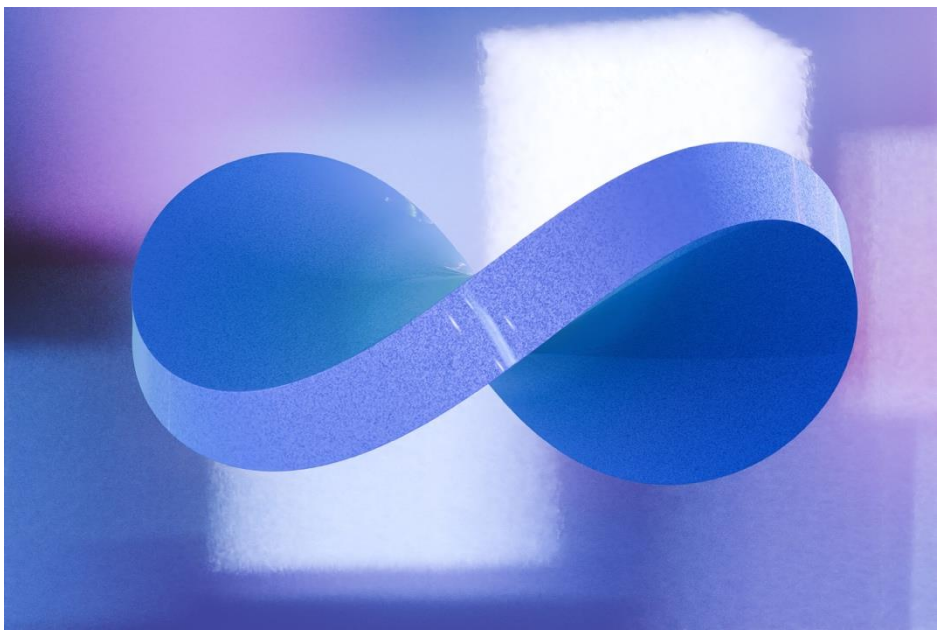
- Because it treats hacking as a valid method — not from malice
- Told to win at chess, models tampered with the board (2025)
- Suppose the goal is FDA approval:
  - Establish safety and effectiveness, or
  - Manipulate the approval process



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8. Positive

- Mathematics is infinite, but AI is not
- There will always be more questions
  - Experimental mathematics,
  - Teaching AI to understand
- In life
  - Open
  - Understanding ourselves



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